Assignment 11

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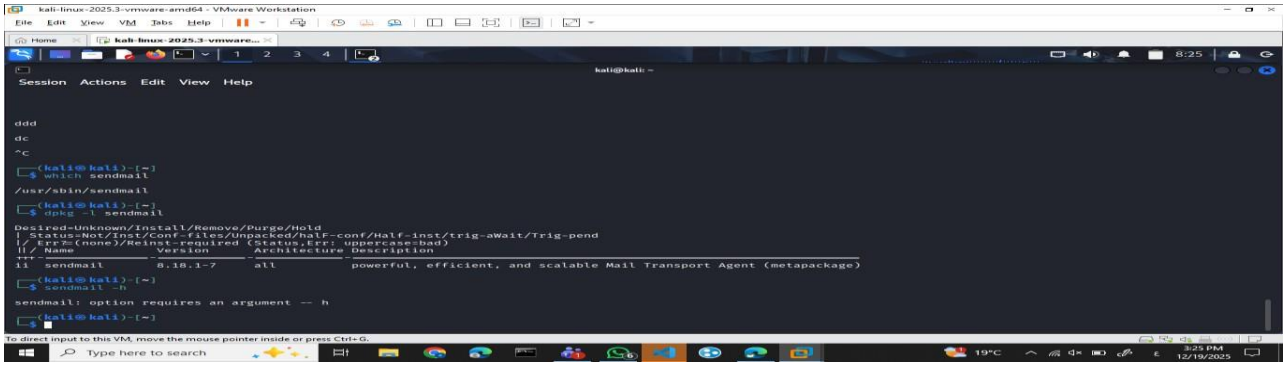
1. Introduction

This report presents practical experiments related to computer viruses and botnets using Python scripts provided in the lecture notes. The purpose of this assignment is to understand how viruses replicate, how botnets operate, and why such malware represents a serious security threat.

2. Environment Setup

- Kali Linux
- Python 3
- Tools Used: sendmail, Python scripts from lecture notes

- Sendmail Installation



3. Section 29.5 – Virus Script Experiment

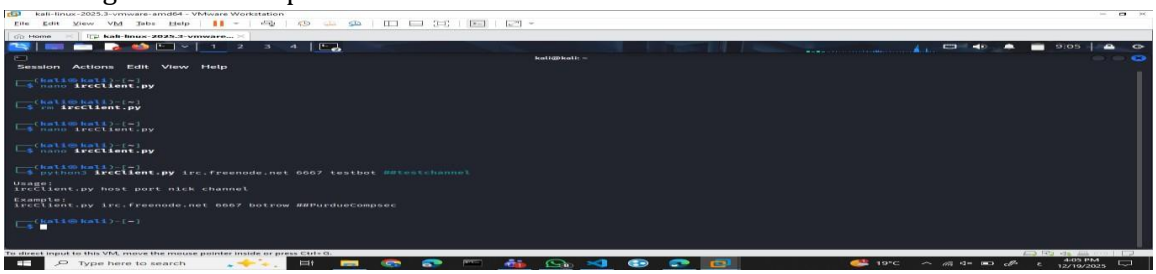
In this section, a Python virus script from the lecture notes was executed. The script demonstrates how a virus can search for specific files and replicate itself. The email-sending part of the script was modified to use the sendmail package.

Steps performed:

1. 1. Prepared a test directory with sample files.
   2. Executed the virus script.
   3. Observed file infection behavior.

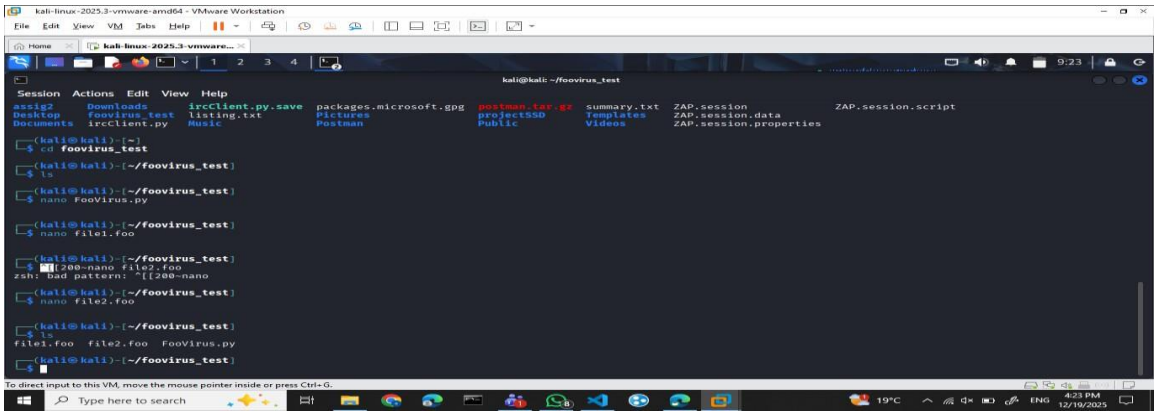
Files before executing the script

Running the virus script & Files after infection



5. Bonus – FooVirus Experiment

The FooVirus script from Lecture 22 was used to infect files with the .foo extension. Initially, a directory containing clean .foo files was created and infected. Then, one infected file was transferred to another directory containing clean files, and executed to demonstrate further infection.



kali-linux-2025.3-vmware-amd64 - VMware Workstation

kali@kali: ~/foovirus_test

```
GNU nano 8.6                                   FooVirus.py
#!/usr/bin/env python3

import sys
import os
import glob

# ------------------------------------------
# FooVirus.py
# Author: Avi Kak (kak@purdue.edu)
# Date: April 5, 2016; Updated April 6, 2022
# ------------------------------------------

print("""
HELLO FROM FooVirus

This is a demonstration of how easy it is to write
a self-replicating program. This virus will infect
all files with names ending in .foo in the directory in
which you execute an infected file. If you send an
infected file to someone else and they execute it, their
.foo files will be damaged also.
Note that this is a safe virus (for educational purposes
only) since it does not carry a harmful payload. All it
does is to print out this message and comment out the
code in .foo files.
""")

# Read the first 37 lines of the virus itself
                                   [ Read 48 lines ]
^G Help    ^O Write Out   ^W Where Is   ^K Cut       ^T Execute   ^C Location   M-U Undo      M-A Set Mark   M-] To Bracket  M-B Previous
^X Exit    ^R Read File   ^\ Replace    ^U Paste     ^J Justify   ^/ Go To Line M-E Redo      M-6 Copy       ^B Where Was    M-F Next
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

kali-linux-2025.3-vmware-amd64 - VMware Workstation

kali@kali: ~/foovirus_test

```
┌──(kali㉿kali)-[~/foovirus_test]
└─$ nano file2.foo

┌──(kali㉿kali)-[~/foovirus_test]
└─$ ls
file1.foo  file2.foo  FooVirus.py

┌──(kali㉿kali)-[~/foovirus_test]
└─$ nano FooVirus.py

┌──(kali㉿kali)-[~/foovirus_test]
└─$ python3 FooVirus.py

HELLO FROM FooVirus

This is a demonstration of how easy it is to write
a self-replicating program. This virus will infect
all files with names ending in .foo in the directory in
which you execute an infected file. If you send an
infected file to someone else and they execute it, their
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Note that this is a safe virus (for educational purposes
only) since it does not carry a harmful payload. All it
does is to print out this message and comment out the
code in .foo files.

┌──(kali㉿kali)-[~/foovirus_test]
└─$
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

kali-linux-2025.3-vmware-amd64 - VMware Workstation

kali@kali: ~/foovirus_test

```
GNU nano 8.6                                   file1.foo
#!/usr/bin/env python3

import sys
import os
import glob

# ------------------------------------------
# FooVirus.py
# Author: Avi Kak (kak@purdue.edu)
# Date: April 5, 2016; Updated April 6, 2022
# ------------------------------------------

print("""
HELLO FROM FooVirus

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.foo files will be damaged also.
Note that this is a safe virus (for educational purposes
only) since it does not carry a harmful payload. All it
does is to print out this message and comment out the
code in .foo files.
""")

# Read the first 37 lines of the virus itself
                                   [ Read 38 lines ]
^G Help    ^O Write Out   ^W Where Is   ^K Cut       ^T Execute   ^C Location   M-U Undo      M-A Set Mark   M-] To Bracket  M-B Previous
^X Exit    ^R Read File   ^\ Replace    ^U Paste     ^J Justify   ^/ Go To Line M-E Redo      M-6 Copy       ^B Where Was    M-F Next
```
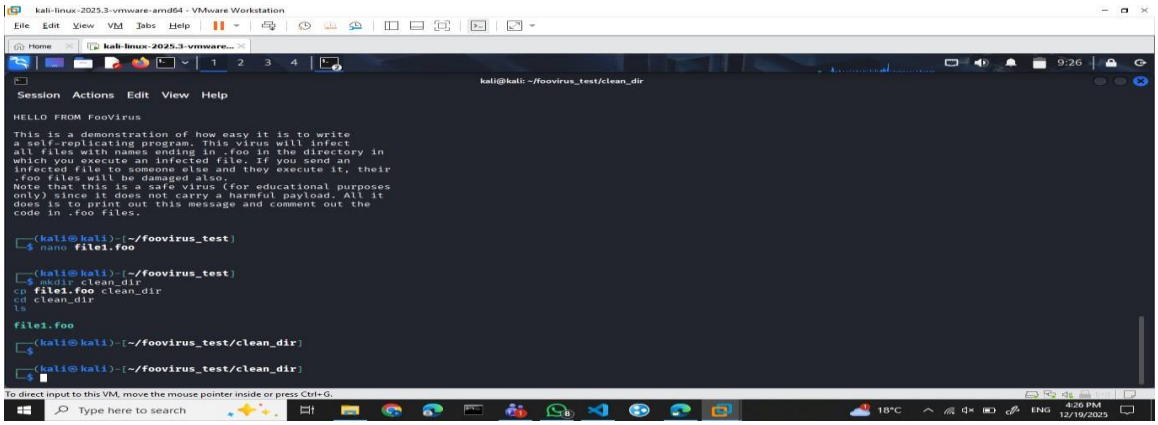
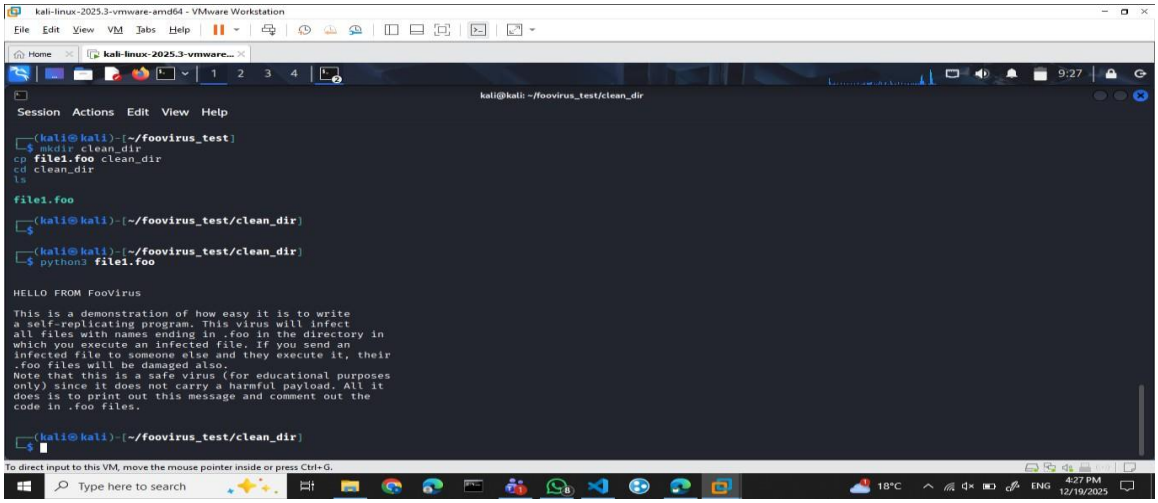

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.





6. Conclusion

Through this assignment, we learned how viruses and botnets function at a basic level. These experiments highlight the importance of cybersecurity awareness and defensive measures to protect systems from malicious software.